

Shot Folder Auto-Creator - Level 3

Objective: Extend your Level-2 tool by adding a **manual logging system** that records every action into a `tool_log.txt` file with timestamps.

In real VFX pipelines, every tool must **log what it does** — not just print to screen — so that production, leads, or TDs can verify actions later.

You have **not** learned the `logging` module yet, so this task will teach you how to build a **manual log writer** using basic file operations.

This means:

- ✓ Every important event = one printed message AND one logged line
 - ✓ Log file **must be persistent** and keep old history (append mode)
-

New Requirements

- 1 Create / Update a log file called:

`tool_log.txt`

- ✓ Must be in the **same folder as your script**
- ✓ Must use **append mode** ("a") — do NOT overwrite previous logs
- ✓ Every event must be logged on its own line

- 2 Log Format (Mandatory)

Each log entry must follow this exact structure:

```
[YYYY-MM-DD HH:MM:SS] MESSAGE
```

Examples:

```
[2025-11-06 13:20:11] CREATED folder: SH030/comp/renders
[2025-11-06 13:20:11] SKIPPED (already exists): SH030/cache
[2025-11-06 13:20:12] ERROR: shot_structure.json not found
[2025-11-06 13:20:14] COMPLETED shot creation: SH030
```

- ✓ Use `datetime.datetime.now()` to generate timestamp
- ✓ Do **not** hard-code timestamps

- 3 What Should Be Logged?

Operation	Example Log Line
Folder created	CREATED folder: SH030/comp
Folder already exists	SKIPPED (already exists): SH030/comp
JSON file missing	ERROR: shot_structure.json not found
Invalid root path	ERROR: Invalid root path
Finished processing	COMPLETED shot creation: SH030

- ✅ If you **print it**, you must also **log it**
- ❌ Do NOT log blank lines or multi-line messages

4 Level-3 Does NOT Change

Feature	Status
sys.argv only (no argparse yet)	✅
JSON-driven folder structure	✅ (same as Level-2)
Nested folder creation	✅
Error handling rules	✅
Command format	✅ (no change)

Example Terminal Output

```
Reading folder structure from shot_structure.json...
Creating shot: SH030
Created folder: SH030/plate
Created folder: SH030/comp
Created folder: SH030/comp/scripts
Created folder: SH030/cache
Shot SH030 created successfully!
```

Example tool_log.txt After Running

```
[2025-11-06 13:20:11] CREATED folder: SH030/plate
[2025-11-06 13:20:11] CREATED folder: SH030/comp
[2025-11-06 13:20:11] CREATED folder: SH030/comp/scripts
[2025-11-06 13:20:11] CREATED folder: SH030/cache
[2025-11-06 13:20:12] COMPLETED shot creation: SH030
```

What You Must Submit

Item	Required
Updated script (<code>shotsetup.py</code>)	
Example <code>tool_log.txt</code> showing logs from at least 2 runs	
Screenshot of console output	
Screenshot of updated folder structure	

Skills You Are Practicing

Area	Skill
Python	File writing (append mode), timestamps, string formatting
CLI	Combining print + log output
Pipeline Thinking	Auditable + traceable tool behavior
Error Thinking	“Did it fail? Did we record it?”
